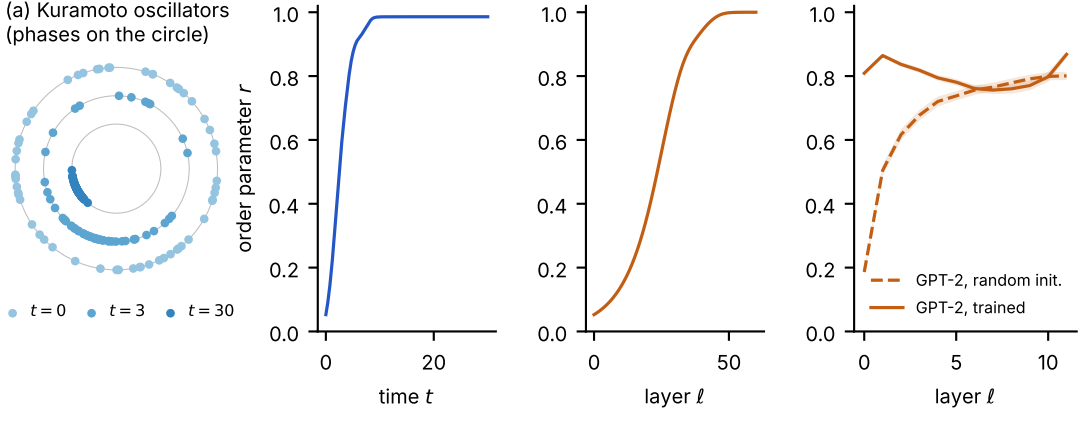
(a) Kuramoto oscillators
(phases on the circle)
t = 0
t = 3
t = 30
(b) synchronization
over time
order parameter r
1.0
0.8
0.6
0.4
0.2
0.0
0
20
time t
(c) idealized attention:
tokens cluster over depth
1.0
0.8
0.6
0.4
0.2
0.0
0
50
layer ℓ
(d) GPT-2 measured:
training resists collapse
1.0
0.8
0.6
0.4
0.2
0.0
GPT-2, random init.
GPT-2, trained
0
5
10
layer ℓ